Inventory Analytics for EV Spare Parts

Group members: [Name 1], [Name 2], [Name 3]
Fill in your team's names here before submitting.

1. Problem Statement and Objective

A regional EV service company operates a central spare-parts warehouse supplying several service centers. Demand for spare parts is uncertain: some parts are ordered routinely, while others are expensive and rarely demanded but critical when needed. Poor inventory planning creates two opposing costs – excess stock ties up capital and incurs holding cost, while insufficient stock causes stockouts, delayed repairs, and lost service revenue.

The warehouse manager wants to move beyond simply forecasting demand accurately, toward understanding the *operational cost* of forecasting errors. This project analyzes 78 weeks of historical demand and inventory data across 24 spare parts, compares several forecasting methods using both statistical error and business cost, and produces concrete reorder recommendations.

Central question: *Can a forecasting method with low statistical error still create high inventory cost?*

Scope: the analysis covers data validation, exploratory analysis of demand and inventory performance, four forecasting methods evaluated via walk-forward validation, a dual (error vs. cost) evaluation, and a reorder-point recommendation system exported as a CSV deliverable.

2. Data and Assumptions

Three weekly tables were used: `parts.csv` (24 parts; cost, price, lead time, holding and stockout cost rates), `demand.csv` (units requested per part per week), and `inventory.csv` (stock on hand and replenishment per part per week, before demand is fulfilled). All three were validated to contain no missing values, no duplicate keys, no date gaps, and no negative values before any further analysis.

After merging demand and inventory on `(week_start, part_id)`, five variables were derived exactly as specified:

- `available_inventory = stock_on_hand + replenishment`
- `units_fulfilled = min(available_inventory, units_demanded)`
- `unmet_demand = max(units_demanded - available_inventory, 0)`

- `ending_stock = max(available_inventory - units_demanded, 0)`
- `stockout = 1` if `unmet_demand > 0`, else 0

Reconciliation (demand = fulfilled + unmet) and the week-to-week stock-chaining property were both verified across the full dataset.

Key assumptions, stated explicitly:

- Forecasts and order quantities are rounded to integer units before any inventory calculation.
- A time-based 75/25 train/test split (58/20 weeks) is used, never shuffled.
- MAPE excludes zero-demand weeks to avoid division by zero.
- Cost model: overage (forecast > actual) is costed at `holding_cost_per_unit_week`; underage (forecast < actual) is costed at `stockout_cost_per_unit`. Each week's forecast is treated as the stocking decision that would have been made.
- Safety stock uses a fixed 95% service level ($z = 1.65$) applied uniformly; a per-part newsvendor critical ratio was tested first but rejected because `stockout_cost_per_unit` exceeds `holding_cost_per_unit_week` by 1–2 orders of magnitude for every part, causing the ratio to saturate near 1.0 and lose any differentiating power.
- A weekly review cycle is assumed, so recommended order quantity brings stock up to the reorder point with no additional review-period buffer.

3. Methodology: Algorithmic Workflow

The analysis proceeds through six phases. The two most decision-heavy components – walk-forward forecasting and reorder classification – are given in pseudocode below; the remaining phases (data audit, merge, EDA, cost aggregation) are straightforward groupby/aggregate operations over the merged weekly table.

```
Algorithm 1:  Walk-forward forecast evaluation
methods = {naive, moving_average, exp_smoothing, holt}
for each part p in 24 parts:
   series = sorted weekly demand for p
   for each test week t (20 weeks, chronological):
      history = series[week < t]   % never includes t or later
      actual = series[t]
      for each method m in methods:
         forecast = round(clip(m(history), 0, ∞))
         record (p, t, m, forecast, actual)
return long table of (part, week, method, forecast, actual)
```

```
Algorithm 2:  Reorder classification (checked in order; first match wins)
function classify(part):
   if current_stock < reorder_point:
```

```
   return "Needs immediate replenishment"
else if stockout_rate > 0.05:
   return "High stockout risk"
else if unit_cost > median_cost and mean_demand < median_demand:
   return "Expensive low-demand -- do not overstock"
else if current_stock > 2 × reorder_point:
   return "Excessive inventory"
else:
   return "Adequate -- no action needed"
```

Demand pattern classification used the Syntetos–Boylan method (ADI and CV^2) rather than an arbitrary zero-demand threshold: ADI $= n/n_0$ (weeks over nonzero-demand weeks) and $CV^2 = (s_0/\bar{x}_0)^2$ over nonzero weeks. All 24 parts classified as *Smooth* (ADI ≈ 1.0, $CV^2 < 0.49$), ruling out Croston's method. Holt's method (trend, no seasonal component – 78 weeks is insufficient for a full seasonal cycle) was kept in the comparison regardless, letting the cost evaluation decide its value empirically.

Safety stock and reorder point:

$$\text{safety\_stock} = \lceil z \cdot \sigma_{\text{demand}} \cdot \sqrt{L} \rceil, \qquad \text{reorder\_point} = \lceil \hat{d} \cdot L + \text{safety\_stock} \rceil$$

where L is lead time in weeks, σ_{demand} is each part's weekly demand standard deviation, and $\hat{d}$ is the forecasted weekly demand from its cost-cheapest method.

4. Findings and Managerial Insights

Error vs. cost, in aggregate: averaged across all 24 parts, Holt's method was both the most accurate (MAE 3.17, MAPE 33.7%) and the cheapest (average total cost $5,714), ahead of exponential smoothing, moving average, and naive (worst on every metric).

The disagreement – the central result: at the level of individual parts, this agreement breaks down. On 11 of 24 parts, the most statistically accurate method was *not* the cheapest one – choosing accuracy over cost on those parts costs **35.8% more on average**, peaking at **152.9% more** on part P023 (Figure 1). A robustness check using RMSE instead of MAE to define "most accurate" gives 9 of 24 parts disagreeing (7 in common with the MAE-based set) – a similarly-sized effect, not an artifact of one specific metric.

Mechanism: the driver is bias direction combined with cost asymmetry, not raw error magnitude. Parts P004 and P008 sit in the expensive/high-demand segment (Figure 3), where stockout cost dominates holding cost by roughly 10x – visible for most parts in Figure 2 – and this asymmetry is the engine of the entire finding: without it, which direction a method's errors lean would not matter for cost at all. On the high-stockout-cost parts, Holt's slightly stronger under-forecasting bias is punished disproportionately by the high per-unit stockout cost, even though its average error (MAE) was smaller than the alternative method's.

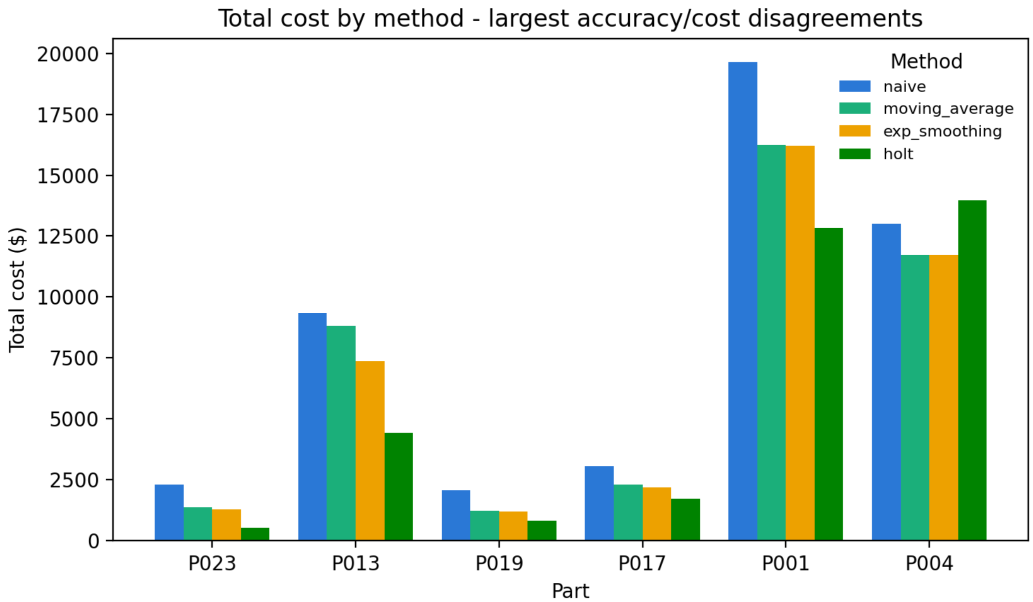

Figure 1: Total cost by method for the parts with the largest accuracy/cost disagreement. Holt (green) is the cheapest bar for five of six parts shown, but the most expensive for P004 – the same method flips from best to worst choice depending on the part's cost structure.

Business recommendations:

1. **Do not select a forecasting method by accuracy alone.** Evaluate cost per part; a globally "best" method (Holt, here) can be the worst choice on specific high-stakes parts.
2. **15 of 24 parts require immediate reordering** per `reorder_recommen- dations.csv`, including three (P002, P006, P016) currently at zero stock on hand.
3. **P012 is a stockout-risk part to monitor**, despite currently adequate stock – it has a 5.1% historical stockout rate.
4. **P001 and P010 are expensive, low-demand parts**: avoid routine over-ordering: their risk is cash tied up in slow-moving inventory, not stockouts.
5. **Limitations:** Holt's method occasionally failed to fully converge on short series (58–77 points); the 95% service level is a fixed assumption, not individually optimized per part; recommended order quantity restores stock only to the reorder point rather than a higher order-up-to target, so a part can re-trigger a reorder soon after; and the reorder policy has not yet been tested against future (out-of-sample) data.

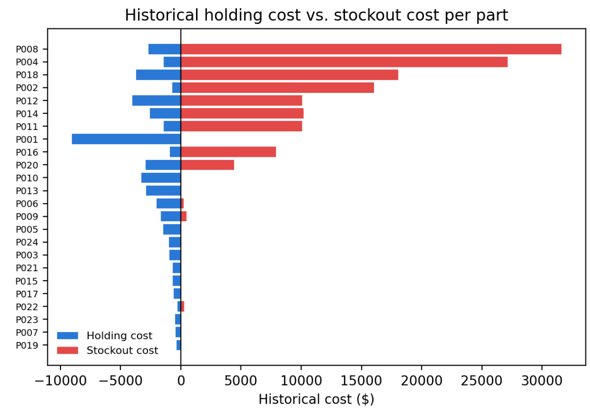

Figure 2: Historical holding cost (blue) vs. stockout cost (red) per part. Stockout cost dominates for most parts – the reason bias direction, not error magnitude, drives cost.

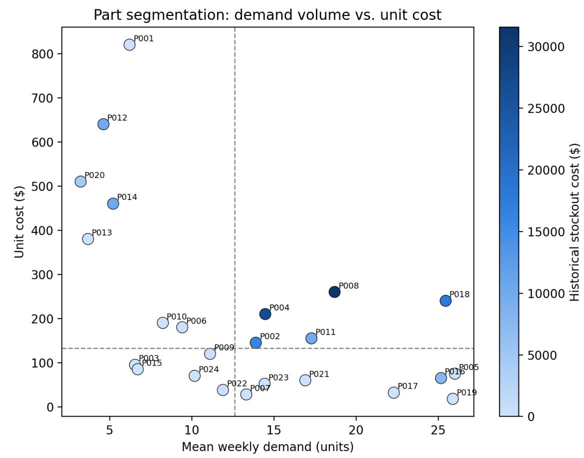

Figure 3: Parts segmented by demand volume and unit cost (median split), colored by historical stockout cost. The expensive/high-demand quadrant carries the largest exposure.

5. Team Member Contributions

Member	Primary responsibilities
Name 1	*e.g., data audit, merge, derived variables; problem formulation*
Name 2	*e.g., EDA, demand classification, segmentation, visualization*
Name 3	*e.g., forecasting methods, walk-forward engine, dual evaluation*

Fill in actual names, roles, and note explicitly if any member did not contribute. Roles may be shared, but each member's main contribution should remain identifiable, and every member should be able to answer questions on any part of the project.

Appendix

A.1 Error and cost metrics by method (averaged across all 24 parts):

Method	MSE	RMSE	MAE	MAPE (%)	Overage $	Total cost $
Holt	17.59	3.88	3.17	33.66	33.07	5,713.59
Exp. smoothing	18.89	4.03	3.26	34.78	32.17	6,598.02
Moving average	20.06	4.15	3.35	35.06	34.00	6,786.27
Naive	31.57	5.23	4.19	43.50	41.82	8,307.97

A.2 Disagreement detail – parts where the most-accurate method differs from the cheapest, sorted by cost premium:

Part	Most accurate	Cheapest	Acc. cost $	Cheap cost $	Premium (%)
P023	moving_average	holt	1,379.0	545.2	152.9
P013	exp_smoothing	holt	7,363.8	4,446.6	65.6
P019	moving_average	holt	1,247.1	836.4	49.1
P017	exp_smoothing	holt	2,195.1	1,739.8	26.2
P001	exp_smoothing	holt	16,205.8	12,845.3	26.2
P004	holt	exp_smoothing	13,962.9	11,734.5	19.0
P008	holt	moving_average	8,849.3	7,653.0	15.6
P021	moving_average	holt	3,379.7	2,999.7	12.7
P007	holt	exp_smoothing	1,556.2	1,401.7	11.0
P010	moving_average	exp_smoothing	5,009.6	4,561.0	9.8
P005	holt	exp_smoothing	5,555.6	5,281.0	5.2

Note: P022 is correctly excluded here. An earlier tie-breaking rule (resolving a 2-way accuracy tie by row order rather than by checking whether any tied method was also cheapest) had misclassified it as disagreeing; the corrected, order-independent rule shows its tied accurate methods include the cheapest one, i.e. it actually agrees.
Average premium across all 11 disagreeing parts: **35.8%**.